

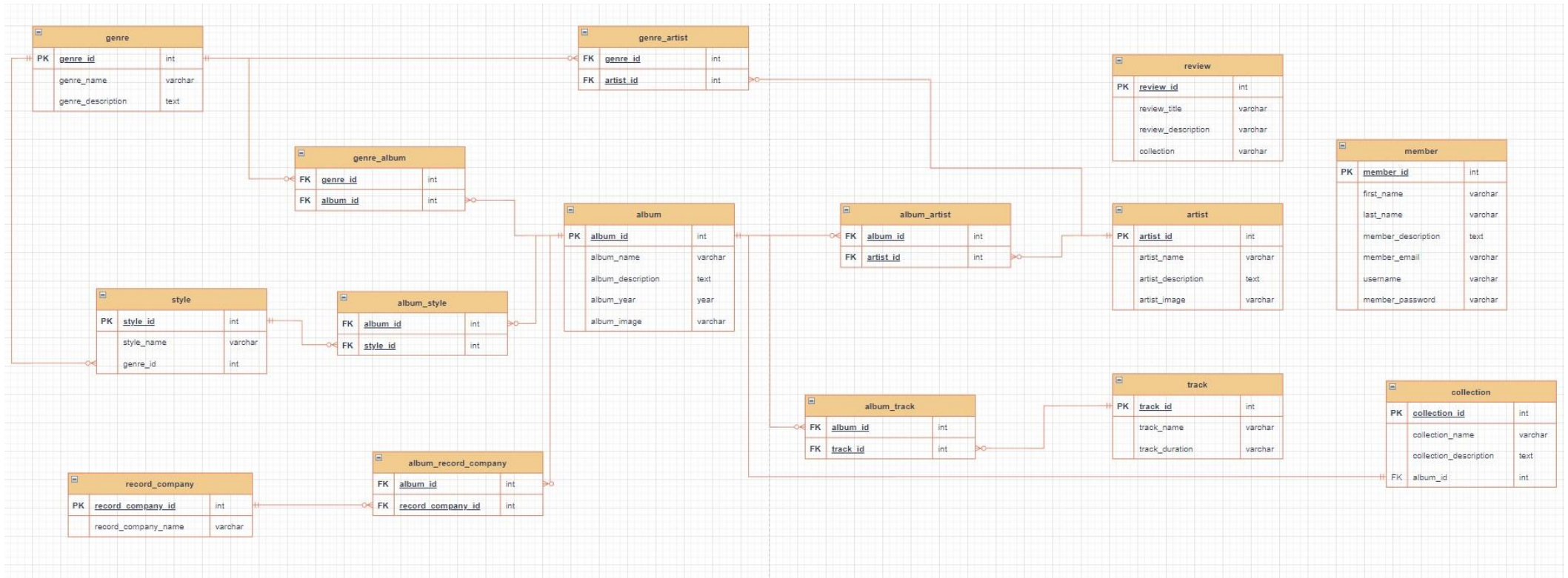


School of Electronics, Electrical
Engineering and Computer Science

CSC7062 Web Development

stacksofwax

Entity Relationship (ER) Diagram



System Development

My approach to this project was as follows:

1. Creation of HTML and CSS templates with the aid of Bootstrap v5.0

I used the Bootstrap framework to aid me in creating a robust, modern and responsive website.

2. Database Design using Drawio

Initial during this stage I used excel to make a note of possible entities and how they relate to each other (e.g., One album can have many artists, therefore this would be a one-to-many and require a bridging table). Once all the relationships between the tables and their entities had been chosen, I drew up the Entity Relationship Diagram using Drawio software.

3. Database creation

At this point I began populating the database with information such as artists and albums. At this stage I used test data for tables such as member/review to ensure that the code was working correctly.

4. Initialisation of Node.js app

Once confident with my database design, and understanding the data that I would be working with and displaying I initialised a Node project. After completing step 5, I took the HTML and CSS I have created and converted these to ejs files.

5. Installation and instantiation of dependencies/required node modules

At this point I ensured I had installed and instantiated the required modules and dependencies I would need to make the project work. This included:

- `const express = require('express');`
- `const path = require('path');`
- `const bodyparser = require('body-parser');`
- `const cookieParser = require('cookie-parser');`
- `const sessions = require('express-session');`
- `const con = require('./connection.js');`

Having the above dependencies ensured I was able to parse form data and carry out other important tasks.

I also at this point in the development of the project, created a connection file that would allow me to pull data from the database.

6. Brief layout of get and post routes required for the project

Page	Route Name	Routes Required
Home Page	/	get
Registration Page	signup	get, post
Log in Page	login	get, post
Artists Page	artist artistinfo	get
Collections	collections createlist	get, post
Albums Page	album	get
Genre Page	genre genreinfo	get
Review	reviewcollection	get, post

7. Development of sql statements required for each route

Once I had the basis of the project set up, I took each route and constructed SQL statements for each that would ensure the right data was pulled/pushed to the database as required.

8. Finalisation and creation of separate routes in order to produce cleaner code

Once happy with the functionality of the web app, I cleaned up my final code by creating separate routes for each of the web pages. I put these under a folder named 'routes' and referenced the particular files in app.js. Unfortunately due to time restraints I was unable to complete this step, however I felt it best that the app ran and was functional.

9. Production of Report and Video Demonstration

Running the System

In order to run this web application on your local machine, you must do the following:

1. Download and unzip the source file
2. Open a text editor such as visual studio code
3. Open the `stacks_of_wax` file on the code editor
4. With the assumption that MAMP/XAMPP, as well as Node.js are installed on your machine open a new terminal within your text editor
5. Run the command `'npm install'`
6. This should pull down the `node_modules` folder, along with the required Javascript files needed for the project based on the dependencies of the web application (These two steps are not necessary as I was unable to delete the node folder due to administration privileges)
7. Ensure that the correct port has been referenced for the database connection depending on whether you are using a mac or windows machine
8. Run the command `'npx nodemon'` in the terminal, this should run the `app.js` file
9. Once running and a connection to the database has been confirmed, you should be able to open the web application on your browser
10. It is recommended that you begin with the homepage by accessing `http://localhost:3000`

URI/References

Video Demonstration URI: <https://youtu.be/cw-9lcRWgqk>

References and Dependencies:

- Bootstrap v5.0 (Javascript/CSS).
- https://www.tutsmake.com/node-js-mysql-autocomplete-search-example/?ref=morioh.com&utm_source=morioh.com – code used and adapted to current project:

```
// Search
app.get('/search', (req, res) => {
  con.query('SELECT artist_name FROM artist WHERE artist_name LIKE "%" +
req.query.term + "%"',
    (err, rows, fields) => {
      if (err) throw err;
      var data = [];
      for (i = 0; i < rows.length; i++) {
        data.push(rows[i].artist_name);
      }
      res.end(JSON.stringify(data));
    });
});
```

